

UNION TOOL

# Tungsten Carbide End Mills UNIMAX Series

HARDMAX 4 Flutes Taper Neck Radius End Mills

Add 28

## HTNRS

Total 111 Models

Vol. 3



UNION TOOL CO.

# HARDMAX 4 Flute Taper Neck Radius End Mills

Size  $\varnothing 1 \sim \varnothing 6$



# HTNRS

Super  
MG

HARD  
MAX

45°

R

R

Shank Dia  
0/-0.005

Back Taper  
Geometry

Variable  
Pitch

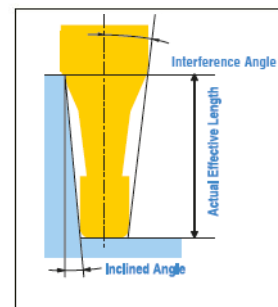
Additional 28 Models

Material Applications (☆ Highly Recommended ◎ Recommended ○ Suggested)

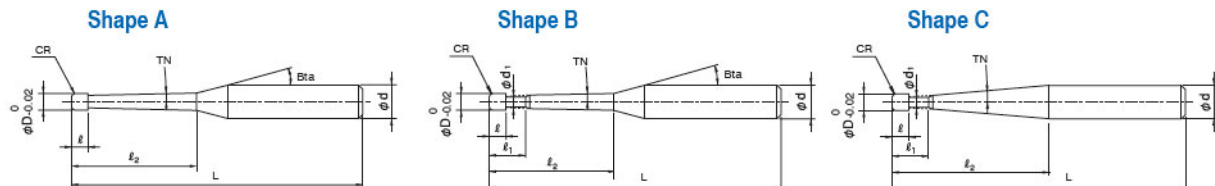
| Work Material                    |                                    |                                     |                 |         |              |           |                    |          |        |          |                             |                    |                             |                     |
|----------------------------------|------------------------------------|-------------------------------------|-----------------|---------|--------------|-----------|--------------------|----------|--------|----------|-----------------------------|--------------------|-----------------------------|---------------------|
| CARBON<br>STEELS<br>S45C<br>S55C | ALLOY<br>STEELS<br>SK / SCM<br>SUS | PREHARDENED<br>STEELS<br>NAK<br>HPM | HARDENED STEELS |         |              | CAST IRON | ALUMINUM<br>ALLOYS | GRAPHITE | COPPER | PLASTICS | GLASS<br>FILLED<br>PLASTICS | TITANIUM<br>ALLOYS | HEAT<br>RESISTANT<br>ALLOYS | CEMENTED<br>CARBIDE |
|                                  |                                    |                                     | ~ 55HRC         | ~ 60HRC | ~ 70HRC      |           |                    |          |        |          |                             |                    |                             |                     |
|                                  |                                    | ◎                                   | ◎               | ◎       | ○<br>~ 65HRC | ○         |                    |          |        |          |                             |                    |                             |                     |

## Features

**4 Flute Taper Neck Radius End Mills for milling hard materials.**  
**Corner radius design from the edge to the periphery ensures less cutting resistance, and the variable pitch design minimizes chattering and vibration.**  
**Can achieve stable milling and excellent surface finish on deep milling.**  
**HARDMAX coating offers longer tool life when milling hard materials.**  
**Recommended to use with any type of coolant.**

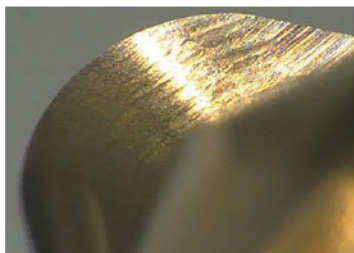


The shank taper angle shown is not an exact value and to avoid contact with the work piece, we recommend the user controls the precise value of this angle. shank taper angle should not make contact with the work piece.



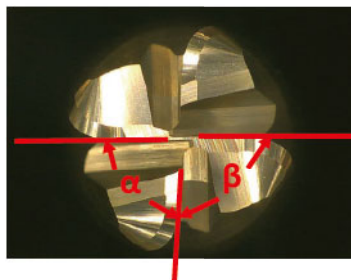
### Feature ①

**Seamless Corner Radius**  
**High rigidity! Less cutting resistance!**



### Feature ②

**Variable Pitch design**  
**Minimizing vibration and chattering !**



Variable Pitch :  $\alpha \neq \beta$

### Feature ③

**A wide choice of Taper Neck Angles available**  
**More efficient with 1.4° 1.9° 2.9°!**



# HARDMAX 4 Flute Taper Neck Radius End Mills

Total 111 models

Unit (mm)

| Model<br>Number    | Outside<br>Diameter<br>Ø D | Corner<br>Radius<br>CR | Neck Taper<br>Angle<br>TN | Neck<br>Length<br>ℓ <sub>2</sub> | Effective<br>Length<br>ℓ <sub>1</sub> | Length<br>of Cut<br>ℓ | Neck<br>Diameter<br>Ø d <sub>1</sub> | Shank Taper<br>Angle<br>Bta | Overall<br>Length<br>L | Shank<br>Diameter<br>Ø d | Shape | Price<br>¥ | Interference<br>Angle | Effective Length by Inclined Angles<br>- : Interference |       |                 |                 |                 |
|--------------------|----------------------------|------------------------|---------------------------|----------------------------------|---------------------------------------|-----------------------|--------------------------------------|-----------------------------|------------------------|--------------------------|-------|------------|-----------------------|---------------------------------------------------------|-------|-----------------|-----------------|-----------------|
|                    |                            |                        |                           |                                  |                                       |                       |                                      |                             |                        |                          |       |            |                       | 30°                                                     | 1°    | 1°30'           | 2°              | 3°              |
|                    |                            |                        |                           |                                  |                                       |                       |                                      |                             |                        |                          |       |            |                       |                                                         |       |                 |                 |                 |
| HTNRS 4010-020608  | 1                          | R0.2                   | 0.4°                      | 6                                | -                                     | 1                     |                                      | 16°                         | 50                     | 4                        | A     | 10,000     | 7.37°                 | 6.56                                                    | 6.92  | 7.20            | 7.45            | 8.00            |
| HTNRS 4010-021008  |                            |                        |                           | 10                               |                                       |                       |                                      |                             | 50                     | 4                        |       | 10,000     | 5.54°                 | 10.61                                                   | 11.12 | 11.50           | 11.89           | 12.77           |
| HTNRS 4010-022008  |                            |                        |                           | 20                               |                                       |                       |                                      |                             | 60                     | 4                        |       | 10,000     | 3.42°                 | 20.73                                                   | 21.52 | 22.24           | 23.00           | 24.71           |
| HTNRS 4010-023008  |                            |                        |                           | 30                               |                                       |                       |                                      |                             | 70                     | 4                        |       | 12,000     | 2.47°                 | 30.83                                                   | 31.91 | 32.97           | 34.11           | No Interference |
| HTNRS 4010-020618  |                            |                        | 0.9°                      | 6                                |                                       |                       |                                      |                             | 50                     | 4                        |       | 10,000     | 7.49°                 | -                                                       | 6.61  | 6.96            | 7.23            | 7.76            |
| HTNRS 4010-021018  |                            |                        |                           | 10                               |                                       |                       |                                      |                             | 50                     | 4                        |       | 10,000     | 5.65°                 | -                                                       | 10.66 | 11.15           | 11.53           | 12.38           |
| HTNRS 4010-021518  |                            |                        |                           | 15                               |                                       |                       |                                      |                             | 50                     | 4                        |       | 10,000     | 4.33°                 | -                                                       | 15.72 | 16.35           | 16.92           | 18.17           |
| HTNRS 4010-022018  |                            |                        |                           | 20                               |                                       |                       |                                      |                             | 60                     | 4                        |       | 10,000     | 3.50°                 | -                                                       | 20.77 | 21.56           | 22.30           | 23.95           |
| HTNRS 4010-022518  |                            |                        |                           | 25                               |                                       |                       |                                      |                             | 60                     | 4                        |       | 10,000     | 2.94°                 | -                                                       | 25.82 | 26.76           | 27.68           | No Interference |
| HTNRS 4010-023018  |                            |                        |                           | 30                               |                                       |                       |                                      |                             | 70                     | 4                        |       | 12,000     | 2.54°                 | -                                                       | 30.87 | 31.96           | 33.06           | No Interference |
| HTNRS 4010-023518  |                            |                        |                           | 35                               |                                       |                       |                                      |                             | 80                     | 4                        |       | 14,000     | 2.23°                 | -                                                       | 35.92 | 37.16           | 38.44           | No Interference |
| HTNRS 4010-024018  |                            |                        |                           | 40                               |                                       |                       |                                      |                             | 80                     | 4                        |       | 14,000     | 1.99°                 | -                                                       | 40.96 | 42.36           | No Interference | No Interference |
| HTNRS 4010-025018  |                            |                        |                           | 50                               |                                       |                       |                                      |                             | 90                     | 4                        |       | 15,000     | 1.64°                 | -                                                       | 51.02 | 52.74           | No Interference | No Interference |
| HTNRS 4010-020628  |                            |                        | 1.4°                      | 6                                |                                       |                       |                                      |                             | 50                     | 4                        |       | 10,000     | 7.61°                 | -                                                       | -     | 6.66            | 7.00            | 7.53            |
| HTNRS 4010-021028  |                            |                        |                           | 10                               |                                       |                       |                                      |                             | 50                     | 4                        |       | 10,000     | 5.77°                 | -                                                       | -     | 10.71           | 11.18           | 12.00           |
| HTNRS 4010-022028  |                            |                        |                           | 20                               |                                       |                       |                                      |                             | 60                     | 4                        |       | 10,000     | 3.59°                 | -                                                       | -     | 20.81           | 21.59           | 23.20           |
| HTNRS 4010-023028  |                            |                        |                           | 30                               |                                       |                       |                                      |                             | 70                     | 4                        |       | 12,000     | 2.61°                 | -                                                       | -     | 30.91           | 32.01           | No Interference |
| HTNRS 40125-020618 | 1.25                       | R0.2                   | 0.9°                      | 6                                | -                                     | 1.25                  | -                                    | 16°                         | 50                     | 4                        | A     | 10,000     | 7.14°                 | -                                                       | 6.63  | 6.97            | 7.24            | 7.77            |
| HTNRS 40125-021018 |                            |                        |                           | 10                               |                                       |                       |                                      |                             | 50                     | 4                        |       | 10,000     | 5.34°                 | -                                                       | 10.68 | 11.16           | 11.55           | 12.40           |
| HTNRS 40125-021518 |                            |                        |                           | 15                               |                                       |                       |                                      |                             | 50                     | 4                        |       | 10,000     | 4.05°                 | -                                                       | 15.74 | 16.37           | 16.93           | 18.18           |
| HTNRS 40125-022018 |                            |                        |                           | 20                               |                                       |                       |                                      |                             | 60                     | 4                        |       | 10,000     | 3.27°                 | -                                                       | 20.79 | 21.57           | 22.31           | 23.96           |
| HTNRS 40125-023018 |                            |                        |                           | 30                               |                                       |                       |                                      |                             | 70                     | 4                        |       | 12,000     | 2.36°                 | -                                                       | 30.89 | 31.97           | 33.07           | No Interference |
| HTNRS 40125-024018 |                            |                        |                           | 40                               |                                       |                       |                                      |                             | 80                     | 4                        |       | 14,000     | 1.84°                 | -                                                       | 40.97 | 42.37           | No Interference | No Interference |
| HTNRS 40125-025018 |                            |                        |                           | 50                               |                                       |                       |                                      |                             | 90                     | 4                        |       | 15,000     | 1.51°                 | -                                                       | 51.03 | 52.75           | No Interference | No Interference |
| HTNRS 4015-030608  | 1.5                        | R0.3                   | 0.4°                      | 6                                | -                                     | 1.5                   | -                                    | 16°                         | 50                     | 4                        | A     | 10,000     | 6.69°                 | 6.62                                                    | 6.96  | 7.23            | 7.47            | 8.01            |
| HTNRS 4015-031008  |                            |                        |                           | 10                               |                                       |                       |                                      |                             | 50                     | 4                        |       | 10,000     | 4.92°                 | 10.66                                                   | 11.15 | 11.52           | 11.91           | 12.79           |
| HTNRS 4015-032008  |                            |                        |                           | 20                               |                                       |                       |                                      |                             | 60                     | 4                        |       | 10,000     | 2.96°                 | 20.78                                                   | 21.55 | 22.26           | 23.03           | No Interference |
| HTNRS 4015-033008  |                            |                        |                           | 30                               |                                       |                       |                                      |                             | 70                     | 4                        |       | 12,000     | 2.12°                 | 30.87                                                   | 31.94 | 33.00           | 34.13           | No Interference |
| HTNRS 4015-030618  |                            |                        | 0.9°                      | 6                                |                                       |                       |                                      |                             | 50                     | 4                        |       | 10,000     | 6.80°                 | -                                                       | 6.69  | 7.01            | 7.27            | 7.79            |
| HTNRS 4015-031018  |                            |                        |                           | 10                               |                                       |                       |                                      |                             | 50                     | 4                        |       | 10,000     | 5.03°                 | -                                                       | 10.73 | 11.19           | 11.57           | 12.42           |
| HTNRS 4015-031518  |                            |                        |                           | 15                               |                                       |                       |                                      |                             | 50                     | 4                        |       | 10,000     | 3.79°                 | -                                                       | 15.79 | 16.39           | 16.95           | 18.20           |
| HTNRS 4015-032018  |                            |                        |                           | 20                               |                                       |                       |                                      |                             | 60                     | 4                        |       | 10,000     | 3.04°                 | -                                                       | 20.84 | 21.60           | 22.34           | 23.99           |
| HTNRS 4015-032518  |                            |                        |                           | 25                               |                                       |                       |                                      |                             | 60                     | 4                        |       | 10,000     | 2.54°                 | -                                                       | 25.88 | 26.80           | 27.72           | No Interference |
| HTNRS 4015-033018  |                            |                        |                           | 30                               |                                       |                       |                                      |                             | 70                     | 4                        |       | 12,000     | 2.18°                 | -                                                       | 30.93 | 32.00           | 33.10           | No Interference |
| HTNRS 4015-034018  |                            |                        |                           | 40                               |                                       |                       |                                      |                             | 80                     | 4                        |       | 14,000     | 1.70°                 | -                                                       | 41.01 | 42.40           | No Interference | No Interference |
| HTNRS 4015-035018  |                            |                        |                           | 50                               |                                       |                       |                                      |                             | 90                     | 4                        |       | 15,000     | 1.39°                 | -                                                       | 51.07 | No Interference | No Interference | No Interference |

\* Additional model

## HARDMAX 4 Flute Taper Neck Radius End Mills

|   | Model Number       | Outside Diameter<br>Ø D | Corner Radius<br>CR | Neck Taper Angle<br>TN | Neck Length<br>ℓ <sub>2</sub> | Effective Length<br>ℓ <sub>1</sub> | Length of Cut<br>ℓ | Neck Diameter<br>Ø d <sub>1</sub> | Shank Taper Angle<br>Bta | Overall Length<br>L | Shank Diameter<br>Ø d | Shape | Price<br>¥ | Interference Angle | Effective Length by Inclined Angles<br>- : Interference |       |                 |                 |                 |
|---|--------------------|-------------------------|---------------------|------------------------|-------------------------------|------------------------------------|--------------------|-----------------------------------|--------------------------|---------------------|-----------------------|-------|------------|--------------------|---------------------------------------------------------|-------|-----------------|-----------------|-----------------|
|   |                    |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    | 30°                                                     | 1°    | 1°30'           | 2°              | 3°              |
|   |                    |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |                 |                 |                 |
| * | HTNRS 4015-030628  | 1.5                     | R0.3                | 1.4°                   | 6                             | 2.7                                | 1.5                | 1.43                              | 16°                      | 50                  | 4                     | B     | 10,000     | 6.92°              | -                                                       | -     | 6.76            | 7.06            | 7.58            |
| * | HTNRS 4015-031028  |                         |                     |                        | 10                            |                                    |                    |                                   |                          | 50                  | 4                     |       | 10,000     | 5.13°              | -                                                       | -     | 10.80           | 11.23           | 12.05           |
| * | HTNRS 4015-032028  |                         |                     |                        | 20                            |                                    |                    |                                   |                          | 60                  | 4                     |       | 10,000     | 3.12°              | -                                                       | -     | 20.89           | 21.65           | 23.25           |
| * | HTNRS 4015-033028  |                         |                     |                        | 30                            |                                    |                    |                                   |                          | 70                  | 4                     |       | 12,000     | 2.24°              | -                                                       | -     | 30.98           | 32.07           | No Interference |
|   | HTNRS 40175-030618 | 1.75                    | R0.3                | 0.9°                   | 6                             | -                                  | 1.75               | -                                 | 16°                      | 50                  | 4                     | A     | 10,000     | 6.37°              | -                                                       | 6.75  | 7.06            | 7.31            | 7.84            |
|   | HTNRS 40175-031018 |                         |                     |                        | 10                            |                                    |                    |                                   |                          | 50                  | 4                     |       | 10,000     | 4.66°              | -                                                       | 10.79 | 11.23           | 11.61           | 12.46           |
|   | HTNRS 40175-031518 |                         |                     |                        | 15                            |                                    |                    |                                   |                          | 60                  | 4                     |       | 10,000     | 3.49°              | -                                                       | 15.84 | 16.43           | 16.99           | 18.24           |
|   | HTNRS 40175-032018 |                         |                     |                        | 20                            |                                    |                    |                                   |                          | 60                  | 4                     |       | 10,000     | 2.78°              | -                                                       | 20.89 | 21.63           | 22.38           | No Interference |
|   | HTNRS 40175-033018 |                         |                     |                        | 30                            |                                    |                    |                                   |                          | 70                  | 4                     |       | 12,000     | 1.99°              | -                                                       | 30.98 | 32.04           | No Interference | No Interference |
|   | HTNRS 40175-034018 |                         |                     |                        | 40                            |                                    |                    |                                   |                          | 80                  | 4                     |       | 14,000     | 1.54°              | -                                                       | 41.06 | 42.44           | No Interference | No Interference |
|   | HTNRS 40175-035018 |                         |                     |                        | 50                            |                                    |                    |                                   |                          | 90                  | 4                     |       | 15,000     | 1.26°              | -                                                       | 51.11 | No Interference | No Interference | No Interference |
|   | HTNRS 4020-052008  | 2                       | R0.5                | 0.4°                   | 20                            | -                                  | 2                  | -                                 | 16°                      | 60                  | 4                     | A     | 11,000     | 2.48°              | 20.86                                                   | 21.60 | 22.30           | 23.06           | No Interference |
|   | HTNRS 4020-052608  |                         |                     |                        | 26                            |                                    |                    |                                   |                          | 60                  | 4                     |       | 11,000     | 1.98°              | 26.92                                                   | 27.83 | 28.75           | No Interference | No Interference |
|   | HTNRS 4020-053008  |                         |                     |                        | 30                            |                                    |                    |                                   |                          | 70                  | 4                     |       | 11,000     | 1.75°              | 30.95                                                   | 31.98 | 33.04           | No Interference | No Interference |
|   | HTNRS 4020-053608  |                         |                     |                        | 36                            |                                    |                    |                                   |                          | 80                  | 4                     |       | 14,000     | 1.49°              | 37.00                                                   | 38.22 | No Interference | No Interference | No Interference |
|   | HTNRS 4020-054008  |                         |                     |                        | 40                            |                                    |                    |                                   |                          | 80                  | 4                     |       | 14,000     | 1.35°              | 41.03                                                   | 42.37 | No Interference | No Interference | No Interference |
|   | HTNRS 4020-051018  |                         |                     | 0.9°                   | 10                            |                                    |                    |                                   |                          | 60                  | 4                     |       | 11,000     | 4.33°              | -                                                       | 10.84 | 11.25           | 11.63           | 12.46           |
|   | HTNRS 4020-051518  |                         |                     |                        | 15                            |                                    |                    |                                   |                          | 60                  | 4                     |       | 11,000     | 3.21°              | -                                                       | 15.88 | 16.45           | 17.01           | 18.25           |
|   | HTNRS 4020-052018  |                         |                     |                        | 20                            |                                    |                    |                                   |                          | 60                  | 4                     |       | 11,000     | 2.54°              | -                                                       | 20.93 | 21.66           | 22.39           | No Interference |
|   | HTNRS 4020-052518  |                         |                     |                        | 25                            |                                    |                    |                                   |                          | 60                  | 4                     |       | 11,000     | 2.11°              | -                                                       | 25.97 | 26.86           | 27.77           | No Interference |
|   | HTNRS 4020-053018  |                         |                     |                        | 30                            |                                    |                    |                                   |                          | 70                  | 4                     |       | 11,000     | 1.80°              | -                                                       | 31.01 | 32.06           | No Interference | No Interference |
|   | HTNRS 4020-053518  |                         |                     |                        | 35                            |                                    |                    |                                   |                          | 80                  | 4                     |       | 14,000     | 1.57°              | -                                                       | 36.05 | 37.26           | No Interference | No Interference |
|   | HTNRS 4020-054018  |                         |                     |                        | 40                            |                                    |                    |                                   |                          | 80                  | 4                     |       | 14,000     | 1.39°              | -                                                       | 41.09 | No Interference | No Interference | No Interference |
|   | HTNRS 4020-054518  |                         |                     |                        | 45                            |                                    |                    |                                   |                          | 90                  | 4                     |       | 15,000     | 1.25°              | -                                                       | 46.10 | No Interference | No Interference | No Interference |
|   | HTNRS 4020-055018  |                         |                     |                        | 50                            |                                    |                    |                                   |                          | 90                  | 4                     |       | 15,000     | 1.14°              | -                                                       | 51.14 | No Interference | No Interference | No Interference |
| * | HTNRS 4020-053028  |                         |                     | 1.4°                   | 30                            |                                    |                    |                                   |                          | 70                  | 4                     | B     | 11,000     | 1.85°              | -                                                       | -     | 31.07           | No Interference | No Interference |
| * | HTNRS 4020-054028  |                         |                     |                        | 40                            |                                    |                    |                                   |                          | 80                  | 4                     |       | 14,000     | 1.43°              | -                                                       | -     | No Interference | No Interference | No Interference |
| * | HTNRS 4020-053038  |                         |                     | 1.9°                   | 30                            |                                    |                    |                                   |                          | 70                  | 6                     |       | 12,000     | 3.39°              | -                                                       | -     | -               | 31.12           | 33.41           |
| * | HTNRS 4020-054038  |                         |                     |                        | 40                            |                                    |                    |                                   |                          | 80                  | 6                     |       | 15,000     | 2.69°              | -                                                       | -     | -               | 41.19           | No Interference |
| * | HTNRS 4020-053058  |                         |                     | 2.9°                   | 30                            |                                    |                    |                                   |                          | 70                  | 6                     | C     | 12,000     | 3.58°              | -                                                       | -     | -               | -               | 31.23           |
| * | HTNRS 4020-054258  |                         |                     |                        | 42                            |                                    |                    |                                   |                          | -                   | 90                    |       | 16,500     | 2.74°              | -                                                       | -     | -               | -               | No Interference |
|   | HTNRS 4030-082008  | 3                       | R0.8                | 0.4°                   | 20                            | -                                  | 3                  | -                                 | 16°                      | 60                  | 6                     | A     | 12,000     | 3.48°              | 20.88                                                   | 21.60 | 22.30           | 23.05           | 24.72           |
|   | HTNRS 4030-082608  |                         |                     |                        | 26                            |                                    |                    |                                   |                          | 60                  | 6                     |       | 12,000     | 2.82°              | 26.94                                                   | 27.84 | 28.74           | 29.72           | No Interference |
|   | HTNRS 4030-083008  |                         |                     |                        | 30                            |                                    |                    |                                   |                          | 70                  | 6                     |       | 12,000     | 2.51°              | 30.97                                                   | 31.99 | 33.04           | 34.16           | No Interference |
|   | HTNRS 4030-083608  |                         |                     |                        | 36                            |                                    |                    |                                   |                          | 80                  | 6                     |       | 14,000     | 2.14°              | 37.02                                                   | 38.22 | 39.48           | 40.82           | No Interference |
|   | HTNRS 4030-084008  |                         |                     |                        | 40                            |                                    |                    |                                   |                          | 80                  | 6                     |       | 14,000     | 1.96°              | 41.05                                                   | 42.38 | 43.78           | No Interference | No Interference |

\* Additional model

## HARDMAX 4 Flute Taper Neck Radius End Mills

| Model Number      | Outside Diameter<br>Ø D | Corner Radius<br>CR | Neck Taper Angle<br>TN | Neck Length<br>ℓ <sub>2</sub> | Effective Length<br>ℓ <sub>1</sub> | Length of Cut<br>ℓ | Neck Diameter<br>Ø d <sub>1</sub> | Shank Taper Angle<br>Bta | Overall Length<br>L | Shank Diameter<br>Ø d | Shape | Price<br>¥ | Interference Angle | Effective Length by Inclined Angles<br>- : Interference |       |        |                 |                 |                 |                 |
|-------------------|-------------------------|---------------------|------------------------|-------------------------------|------------------------------------|--------------------|-----------------------------------|--------------------------|---------------------|-----------------------|-------|------------|--------------------|---------------------------------------------------------|-------|--------|-----------------|-----------------|-----------------|-----------------|
|                   |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    | 30'                                                     | 1°    | 1°30'  | 2°              | 3°              |                 |                 |
|                   |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4030-082018 | 3                       | R0.8                | 0.9°                   | 20                            | -                                  | 3                  |                                   | 16°                      | 60                  | 6                     | A     | 12,000     | 3.56°              | -                                                       | 20.98 | 21.69  | 22.41           | 24.03           |                 |                 |
| HTNRS 4030-082518 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4030-083018 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4030-083518 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4030-084018 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4030-085018 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4030-086018 |                         |                     |                        |                               |                                    |                    | 4.5                               | 2.89                     |                     | 100                   | 6     | B          | 16,000             | 1.40°                                                   | -     | 61.25  | No Interference | No Interference | No Interference |                 |
| HTNRS 4030-083028 |                         |                     | 1.4°                   | 30                            | 70                                 | 6                  |                                   |                          |                     | 12,000                | 2.64° |            | -                  | -                                                       | 31.14 | 32.19  | No Interference | No Interference |                 |                 |
| HTNRS 4030-084028 |                         |                     |                        | 40                            | 80                                 | 6                  |                                   |                          |                     | 14,000                | 2.06° |            | -                  | -                                                       | 41.21 | 42.61  | No Interference | No Interference |                 |                 |
| HTNRS 4030-083038 |                         |                     | 1.9°                   | 30                            | 70                                 | 6                  |                                   |                          |                     | 12,000                | 2.71° |            | -                  | -                                                       | -     | 31.21  | No Interference | No Interference |                 |                 |
| HTNRS 4030-084038 |                         |                     |                        | 40                            | 80                                 | 6                  |                                   |                          |                     | 14,000                | 2.12° |            | -                  | -                                                       | -     | 41.28  | No Interference | No Interference |                 |                 |
| HTNRS 4030-083358 |                         |                     | 2.9°                   | 33                            | -                                  | 80                 |                                   |                          |                     | 6                     | C     |            | 16,500             | 2.64°                                                   | -     | -      | -               | -               | No Interference | No Interference |
| HTNRS 4040-102508 | 4                       | R1                  | 0.4°                   | 25                            |                                    | 4                  |                                   | 16°                      | 60                  | 6                     | A     | 12,000     | 2.12°              | 25.49                                                   | 26.28 | 27.13  | 28.04           | No Interference |                 |                 |
| HTNRS 4040-103008 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4040-103508 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4040-104008 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4040-104508 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4040-105008 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4040-102018 |                         |                     | 0.9°                   | 20                            | -                                  | 4                  |                                   | 16°                      | 60                  | 6                     | A     | 12,000     | 2.64°              | -                                                       | 20.57 | 21.23  | 21.93           | No Interference |                 |                 |
| HTNRS 4040-102518 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4040-103018 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4040-103518 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4040-104018 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4040-105018 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4040-106018 |                         |                     |                        | 60                            |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4040-104928 |                         |                     | 1.4°                   | 49                            | 6                                  | 3.8                |                                   | 16°                      | -                   | 90                    | 6     | C          | 15,000             | 1.21°                                                   | -     | -      | No Interference | No Interference | No Interference |                 |
| HTNRS 4040-106028 |                         |                     |                        | 60                            |                                    |                    |                                   |                          | 100                 | 8                     | B     | 17,000     | 1.88°              | -                                                       | -     | 60.94  | No Interference | No Interference |                 |                 |
| HTNRS 4040-103038 |                         |                     | 1.9°                   | 30                            |                                    |                    |                                   |                          | 70                  | 8                     |       | 16,000     | 3.46°              | -                                                       | -     | -      | 30.89           | 33.13           |                 |                 |
| HTNRS 4040-106738 |                         |                     |                        | 67                            |                                    |                    |                                   |                          | -                   | 120                   | 8     | C          | 23,000             | 1.75°                                                   | -     | -      | -               | No Interference | No Interference |                 |
| HTNRS 4040-104558 |                         |                     | 2.9°                   | 45                            |                                    |                    |                                   |                          |                     | 90                    | 8     |            | 18,000             | 2.62°                                                   | -     | -      | -               | -               | No Interference | No Interference |
| HTNRS 4060-152018 | 6                       | R1.5                | 0.9°                   | 20                            |                                    |                    |                                   |                          | -                   | 6                     | -     | 16°        | 60                 | 8                                                       | A     | 17,000 | 2.69°           | -               | 20.63           | 21.28           |
| HTNRS 4060-153018 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4060-154018 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4060-155018 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |
| HTNRS 4060-156018 |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    |                                                         |       |        |                 |                 |                 |                 |

\* Additional model



## HARDMAX 4 Flute Taper Neck Radius End Mills

| Model Number        | Outside Diameter<br>Ø D | Corner Radius<br>CR | Neck Taper Angle<br>TN | Neck Length<br>ℓ <sub>2</sub> | Effective Length<br>ℓ <sub>1</sub> | Length of Cut<br>ℓ | Neck Diameter<br>Ø d <sub>1</sub> | Shank Taper Angle<br>Bta | Overall Length<br>L | Shank Diameter<br>Ø d | Shape | Price<br>¥ | Interference Angle | Effective Length by Inclined Angles<br>- : Interference |    |                 |                 |                 |
|---------------------|-------------------------|---------------------|------------------------|-------------------------------|------------------------------------|--------------------|-----------------------------------|--------------------------|---------------------|-----------------------|-------|------------|--------------------|---------------------------------------------------------|----|-----------------|-----------------|-----------------|
|                     |                         |                     |                        |                               |                                    |                    |                                   |                          |                     |                       |       |            |                    | 30°                                                     | 1° | 1°30'           | 2°              | 3°              |
| * HTNRS 4060-155128 | 6                       | R1.5                | 1.4°                   | 51                            | 9                                  | 6                  | 5.8                               | -                        | 90                  | 8                     | C     | 17,000     | 1.18°              | -                                                       | -  | No Interference | No Interference | No Interference |
| * HTNRS 4060-153938 |                         |                     | 1.9°                   | 39                            |                                    |                    |                                   |                          | 80                  | 8                     |       | 17,000     | 1.55°              | -                                                       | -  | -               | No Interference | No Interference |
| * HTNRS 4060-156938 |                         |                     | 1.9°                   | 69                            |                                    |                    |                                   |                          | 110                 | 10                    |       | 23,500     | 1.71°              | -                                                       | -  | -               | No Interference | No Interference |
| * HTNRS 4060-154758 |                         |                     | 2.9°                   | 47                            |                                    |                    |                                   |                          | 90                  | 10                    |       | 23,000     | 2.53°              | -                                                       | -  | -               | -               | No Interference |

\* Additional model

# HTNRS Milling Conditions

| WORK MATERIAL |                       |                    |                  |                  | PREHARDENED STEELS / HARDENED STEELS<br>NAK / SKD<br>(30~45HRC) |                    |                                 |                                  | HARDENED STEELS<br>SKD / SKT<br>(45~55HRC) |                    |                                 |                                  | HARDENED STEELS<br>SKD / SKH<br>(55~65HRC) |                    |                                 |                                  |
|---------------|-----------------------|--------------------|------------------|------------------|-----------------------------------------------------------------|--------------------|---------------------------------|----------------------------------|--------------------------------------------|--------------------|---------------------------------|----------------------------------|--------------------------------------------|--------------------|---------------------------------|----------------------------------|
| Model Number  | Outside Diameter (mm) | Corner Radius (mm) | Neck Taper Angle | Neck Length (mm) | Spindle Speed (min <sup>-1</sup> )                              | Feed Rate (mm/min) | a <sub>p</sub> Axial Depth (mm) | a <sub>e</sub> Radial Depth (mm) | Spindle Speed (min <sup>-1</sup> )         | Feed Rate (mm/min) | a <sub>p</sub> Axial Depth (mm) | a <sub>e</sub> Radial Depth (mm) | Spindle Speed (min <sup>-1</sup> )         | Feed Rate (mm/min) | a <sub>p</sub> Axial Depth (mm) | a <sub>e</sub> Radial Depth (mm) |
| 4010-020608   | 1                     | R0.2               | 0.4°             | 6                | 20,000                                                          | 2,600              | 0.06                            | 0.44                             | 11,600                                     | 980                | 0.029                           | 0.23                             | 8,900                                      | 530                | 0.012                           | 0.09                             |
| 4010-021008   |                       |                    |                  | 10               | 19,000                                                          | 2,450              | 0.03                            | 0.42                             | 11,000                                     | 920                | 0.015                           | 0.21                             | 8,500                                      | 480                | 0.008                           | 0.07                             |
| 4010-022008   |                       |                    |                  | 20               | 17,000                                                          | 2,150              | 0.008                           | 0.37                             | 9,700                                      | 800                | 0.005                           | 0.18                             | 7,600                                      | 400                | 0.004                           | 0.05                             |
| 4010-023008   |                       |                    |                  | 30               | 12,000                                                          | 1,500              | 0.003                           | 0.26                             | 7,200                                      | 570                | 0.002                           | 0.12                             | 6,000                                      | 310                | 0.002                           | 0.04                             |
| 4010-020618   |                       |                    | 0.9°             | 6                | 20,000                                                          | 2,600              | 0.06                            | 0.45                             | 11,600                                     | 980                | 0.029                           | 0.24                             | 8,900                                      | 530                | 0.012                           | 0.1                              |
| 4010-021018   |                       |                    |                  | 10               | 19,000                                                          | 2,450              | 0.03                            | 0.43                             | 11,000                                     | 920                | 0.015                           | 0.22                             | 8,500                                      | 480                | 0.008                           | 0.075                            |
| 4010-021518   |                       |                    |                  | 15               | 18,500                                                          | 2,400              | 0.02                            | 0.41                             | 10,700                                     | 880                | 0.01                            | 0.21                             | 8,200                                      | 450                | 0.006                           | 0.065                            |
| 4010-022018   |                       |                    |                  | 20               | 18,000                                                          | 2,300              | 0.01                            | 0.4                              | 10,400                                     | 850                | 0.006                           | 0.2                              | 8,000                                      | 430                | 0.005                           | 0.05                             |
| 4010-022518   |                       |                    |                  | 25               | 17,000                                                          | 2,150              | 0.008                           | 0.38                             | 9,900                                      | 800                | 0.005                           | 0.19                             | 7,700                                      | 410                | 0.004                           | 0.05                             |
| 4010-023018   |                       |                    |                  | 30               | 16,000                                                          | 2,000              | 0.007                           | 0.35                             | 9,400                                      | 750                | 0.004                           | 0.18                             | 7,400                                      | 390                | 0.004                           | 0.05                             |
| 4010-023518   |                       |                    |                  | 35               | 15,000                                                          | 1,850              | 0.006                           | 0.32                             | 8,800                                      | 700                | 0.004                           | 0.16                             | 7,000                                      | 370                | 0.003                           | 0.05                             |
| 4010-024018   |                       |                    |                  | 40               | 14,000                                                          | 1,750              | 0.005                           | 0.3                              | 8,300                                      | 660                | 0.003                           | 0.15                             | 6,700                                      | 350                | 0.003                           | 0.05                             |
| 4010-025018   |                       |                    |                  | 50               | 12,000                                                          | 1,500              | 0.003                           | 0.28                             | 7,200                                      | 570                | 0.002                           | 0.14                             | 6,000                                      | 310                | 0.002                           | 0.05                             |
| 4010-020628   |                       |                    | 1.4°             | 6                | 20,000                                                          | 2,600              | 0.06                            | 0.46                             | 11,600                                     | 980                | 0.029                           | 0.25                             | 8,900                                      | 530                | 0.012                           | 0.11                             |
| 4010-021028   |                       |                    |                  | 10               | 20,000                                                          | 2,600              | 0.04                            | 0.45                             | 11,400                                     | 960                | 0.02                            | 0.24                             | 8,750                                      | 510                | 0.01                            | 0.08                             |
| 4010-022028   |                       |                    |                  | 20               | 19,000                                                          | 2,400              | 0.02                            | 0.4                              | 10,900                                     | 900                | 0.01                            | 0.2                              | 8,400                                      | 470                | 0.005                           | 0.06                             |
| 4010-023028   |                       |                    |                  | 30               | 18,000                                                          | 2,300              | 0.01                            | 0.4                              | 10,400                                     | 850                | 0.006                           | 0.2                              | 8,000                                      | 430                | 0.005                           | 0.05                             |
| 40125-020618  | 1.25                  | R0.2               | 0.9°             | 6                | 16,000                                                          | 2,600              | 0.075                           | 0.56                             | 9,200                                      | 990                | 0.036                           | 0.3                              | 7,100                                      | 540                | 0.015                           | 0.12                             |
| 40125-021018  |                       |                    |                  | 10               | 16,000                                                          | 2,600              | 0.057                           | 0.55                             | 9,200                                      | 990                | 0.027                           | 0.29                             | 7,100                                      | 540                | 0.012                           | 0.1                              |
| 40125-021518  |                       |                    |                  | 15               | 15,500                                                          | 2,500              | 0.04                            | 0.53                             | 8,900                                      | 950                | 0.019                           | 0.27                             | 6,900                                      | 500                | 0.01                            | 0.08                             |
| 40125-022018  |                       |                    |                  | 20               | 15,000                                                          | 2,400              | 0.022                           | 0.51                             | 8,700                                      | 900                | 0.011                           | 0.26                             | 6,700                                      | 470                | 0.007                           | 0.07                             |
| 40125-023018  |                       |                    |                  | 30               | 14,400                                                          | 2,300              | 0.011                           | 0.47                             | 8,300                                      | 860                | 0.006                           | 0.23                             | 6,400                                      | 440                | 0.005                           | 0.06                             |
| 40125-024018  |                       |                    |                  | 40               | 12,800                                                          | 2,000              | 0.008                           | 0.42                             | 7,500                                      | 750                | 0.004                           | 0.21                             | 5,900                                      | 390                | 0.004                           | 0.06                             |
| 40125-025018  |                       |                    |                  | 50               | 11,000                                                          | 1,700              | 0.006                           | 0.37                             | 6,500                                      | 650                | 0.003                           | 0.19                             | 5,300                                      | 350                | 0.003                           | 0.06                             |
| 4015-030608   | 1.5                   | R0.3               | 0.4°             | 6                | 13,500                                                          | 2,600              | 0.09                            | 0.67                             | 7,800                                      | 990                | 0.043                           | 0.36                             | 6,000                                      | 540                | 0.018                           | 0.15                             |
| 4015-031008   |                       |                    |                  | 10               | 13,500                                                          | 2,600              | 0.083                           | 0.66                             | 7,700                                      | 980                | 0.04                            | 0.35                             | 6,000                                      | 540                | 0.017                           | 0.13                             |
| 4015-032008   |                       |                    |                  | 20               | 12,500                                                          | 2,400              | 0.028                           | 0.61                             | 7,000                                      | 880                | 0.015                           | 0.31                             | 5,500                                      | 460                | 0.009                           | 0.09                             |
| 4015-033008   |                       |                    |                  | 30               | 12,000                                                          | 2,300              | 0.012                           | 0.55                             | 6,900                                      | 860                | 0.007                           | 0.27                             | 5,350                                      | 440                | 0.006                           | 0.07                             |
| 4015-030618   |                       |                    | 0.9°             | 6                | 13,500                                                          | 2,600              | 0.09                            | 0.67                             | 7,800                                      | 990                | 0.043                           | 0.36                             | 6,000                                      | 540                | 0.018                           | 0.15                             |
| 4015-031018   |                       |                    |                  | 10               | 13,500                                                          | 2,600              | 0.083                           | 0.67                             | 7,800                                      | 990                | 0.04                            | 0.36                             | 6,000                                      | 540                | 0.017                           | 0.14                             |
| 4015-031518   |                       |                    |                  | 15               | 13,000                                                          | 2,500              | 0.055                           | 0.65                             | 7,500                                      | 950                | 0.029                           | 0.34                             | 5,800                                      | 500                | 0.013                           | 0.12                             |
| 4015-032018   |                       |                    |                  | 20               | 12,500                                                          | 2,400              | 0.035                           | 0.63                             | 7,200                                      | 900                | 0.018                           | 0.32                             | 5,600                                      | 470                | 0.01                            | 0.1                              |
| 4015-032518   |                       |                    |                  | 25               | 12,500                                                          | 2,400              | 0.025                           | 0.61                             | 7,000                                      | 880                | 0.013                           | 0.31                             | 5,500                                      | 460                | 0.009                           | 0.08                             |
| 4015-033018   |                       |                    |                  | 30               | 12,000                                                          | 2,300              | 0.015                           | 0.6                              | 6,900                                      | 860                | 0.008                           | 0.3                              | 5,350                                      | 440                | 0.007                           | 0.07                             |
| 4015-034018   |                       |                    |                  | 40               | 12,000                                                          | 2,300              | 0.012                           | 0.55                             | 6,900                                      | 860                | 0.007                           | 0.27                             | 5,350                                      | 440                | 0.006                           | 0.07                             |
| 4015-035018   |                       |                    |                  | 50               | 10,500                                                          | 2,000              | 0.009                           | 0.5                              | 6,100                                      | 740                | 0.005                           | 0.25                             | 4,850                                      | 380                | 0.005                           | 0.07                             |

# HTNRS Milling Conditions

| WORK MATERIAL |                       |                    |                  |                  | PREHARDENED STEELS / HARDENED STEELS<br>NAK / SKD<br>(30~45HRC) |                    |                                 |                                  | HARDENED STEELS<br>SKD / SKT<br>(45~55HRC) |                    |                                 |                                  | HARDENED STEELS<br>SKD / SKH<br>(55~65HRC) |                    |                                 |                                  |
|---------------|-----------------------|--------------------|------------------|------------------|-----------------------------------------------------------------|--------------------|---------------------------------|----------------------------------|--------------------------------------------|--------------------|---------------------------------|----------------------------------|--------------------------------------------|--------------------|---------------------------------|----------------------------------|
| Model Number  | Outside Diameter (mm) | Corner Radius (mm) | Neck Taper Angle | Neck Length (mm) | Spindle Speed (min <sup>-1</sup> )                              | Feed Rate (mm/min) | a <sub>p</sub> Axial Depth (mm) | a <sub>e</sub> Radial Depth (mm) | Spindle Speed (min <sup>-1</sup> )         | Feed Rate (mm/min) | a <sub>p</sub> Axial Depth (mm) | a <sub>e</sub> Radial Depth (mm) | Spindle Speed (min <sup>-1</sup> )         | Feed Rate (mm/min) | a <sub>p</sub> Axial Depth (mm) | a <sub>e</sub> Radial Depth (mm) |
| 4015-030628   | 1.5                   | R0.3               | 1.4°             | 6                | 13,500                                                          | 2,600              | 0.09                            | 0.67                             | 7,800                                      | 990                | 0.043                           | 0.36                             | 6,000                                      | 540                | 0.018                           | 0.15                             |
| 4015-031028   |                       |                    |                  | 10               | 13,500                                                          | 2,600              | 0.085                           | 0.67                             | 7,800                                      | 990                | 0.04                            | 0.36                             | 6,000                                      | 540                | 0.017                           | 0.15                             |
| 4015-032028   |                       |                    |                  | 20               | 13,000                                                          | 2,500              | 0.05                            | 0.63                             | 7,300                                      | 920                | 0.02                            | 0.33                             | 5,600                                      | 480                | 0.011                           | 0.11                             |
| 4015-033028   |                       |                    |                  | 30               | 12,500                                                          | 2,400              | 0.025                           | 0.61                             | 7,000                                      | 880                | 0.013                           | 0.31                             | 5,500                                      | 460                | 0.009                           | 0.08                             |
| 40175-030618  | 1.75                  | R0.3               | 0.9°             | 6                | 11,500                                                          | 2,600              | 0.105                           | 0.78                             | 6,600                                      | 990                | 0.05                            | 0.42                             | 5,100                                      | 540                | 0.021                           | 0.17                             |
| 40175-031018  |                       |                    |                  | 10               | 11,500                                                          | 2,600              | 0.105                           | 0.78                             | 6,600                                      | 990                | 0.05                            | 0.42                             | 5,100                                      | 540                | 0.021                           | 0.17                             |
| 40175-031518  |                       |                    |                  | 15               | 11,500                                                          | 2,600              | 0.07                            | 0.76                             | 6,500                                      | 950                | 0.037                           | 0.4                              | 5,000                                      | 510                | 0.017                           | 0.14                             |
| 40175-032018  |                       |                    |                  | 20               | 11,000                                                          | 2,450              | 0.047                           | 0.74                             | 6,400                                      | 920                | 0.024                           | 0.38                             | 4,900                                      | 480                | 0.013                           | 0.12                             |
| 40175-033018  |                       |                    |                  | 30               | 11,000                                                          | 2,450              | 0.027                           | 0.71                             | 6,400                                      | 920                | 0.014                           | 0.36                             | 4,900                                      | 480                | 0.01                            | 0.1                              |
| 40175-034018  |                       |                    |                  | 40               | 10,000                                                          | 2,200              | 0.016                           | 0.67                             | 5,800                                      | 820                | 0.009                           | 0.33                             | 4,450                                      | 420                | 0.008                           | 0.08                             |
| 40175-035018  |                       |                    |                  | 50               | 10,000                                                          | 2,200              | 0.013                           | 0.62                             | 5,800                                      | 820                | 0.008                           | 0.31                             | 4,450                                      | 420                | 0.007                           | 0.08                             |
| 4020-052008   | 2                     | R0.5               | 0.4°             | 20               | 9,500                                                           | 2,450              | 0.06                            | 0.85                             | 5,500                                      | 920                | 0.025                           | 0.43                             | 4,250                                      | 480                | 0.015                           | 0.13                             |
| 4020-052608   |                       |                    |                  | 26               | 9,500                                                           | 2,450              | 0.04                            | 0.83                             | 5,500                                      | 920                | 0.021                           | 0.42                             | 4,250                                      | 480                | 0.013                           | 0.12                             |
| 4020-053008   |                       |                    |                  | 30               | 9,000                                                           | 2,300              | 0.03                            | 0.79                             | 5,400                                      | 880                | 0.016                           | 0.41                             | 4,100                                      | 450                | 0.012                           | 0.11                             |
| 4020-053608   |                       |                    |                  | 36               | 9,000                                                           | 2,300              | 0.02                            | 0.75                             | 5,200                                      | 850                | 0.011                           | 0.39                             | 4,000                                      | 430                | 0.01                            | 0.1                              |
| 4020-054008   |                       |                    |                  | 40               | 9,000                                                           | 2,300              | 0.02                            | 0.7                              | 5,200                                      | 850                | 0.01                            | 0.38                             | 4,000                                      | 430                | 0.009                           | 0.1                              |
| 4020-051018   |                       |                    | 0.9°             | 10               | 10,000                                                          | 2,600              | 0.12                            | 0.9                              | 5,800                                      | 990                | 0.057                           | 0.49                             | 4,450                                      | 540                | 0.024                           | 0.2                              |
| 4020-051518   |                       |                    |                  | 15               | 10,000                                                          | 2,600              | 0.09                            | 0.88                             | 5,600                                      | 950                | 0.044                           | 0.47                             | 4,350                                      | 510                | 0.02                            | 0.17                             |
| 4020-052018   |                       |                    |                  | 20               | 9,500                                                           | 2,450              | 0.06                            | 0.86                             | 5,500                                      | 920                | 0.03                            | 0.45                             | 4,250                                      | 480                | 0.016                           | 0.15                             |
| 4020-052518   |                       |                    |                  | 25               | 9,500                                                           | 2,450              | 0.05                            | 0.85                             | 5,500                                      | 920                | 0.025                           | 0.43                             | 4,250                                      | 480                | 0.015                           | 0.13                             |
| 4020-053018   |                       |                    |                  | 30               | 9,500                                                           | 2,450              | 0.04                            | 0.83                             | 5,500                                      | 920                | 0.021                           | 0.42                             | 4,250                                      | 480                | 0.013                           | 0.12                             |
| 4020-053518   |                       |                    |                  | 35               | 9,000                                                           | 2,300              | 0.03                            | 0.81                             | 5,300                                      | 880                | 0.016                           | 0.41                             | 4,100                                      | 450                | 0.012                           | 0.11                             |
| 4020-054018   |                       |                    |                  | 40               | 9,000                                                           | 2,300              | 0.02                            | 0.8                              | 5,200                                      | 850                | 0.012                           | 0.4                              | 4,000                                      | 430                | 0.01                            | 0.1                              |
| 4020-054518   |                       |                    |                  | 45               | 9,000                                                           | 2,300              | 0.02                            | 0.75                             | 5,200                                      | 850                | 0.011                           | 0.39                             | 4,000                                      | 430                | 0.01                            | 0.1                              |
| 4020-055018   |                       |                    |                  | 50               | 9,000                                                           | 2,300              | 0.017                           | 0.75                             | 5,200                                      | 850                | 0.01                            | 0.38                             | 4,000                                      | 430                | 0.009                           | 0.1                              |
| 4020-053028   |                       |                    | 1.4°             | 30               | 9,500                                                           | 2,450              | 0.05                            | 0.85                             | 5,500                                      | 920                | 0.025                           | 0.43                             | 4,250                                      | 480                | 0.015                           | 0.13                             |
| 4020-054028   |                       |                    |                  | 40               | 9,500                                                           | 2,450              | 0.04                            | 0.83                             | 5,500                                      | 920                | 0.02                            | 0.42                             | 4,250                                      | 480                | 0.013                           | 0.12                             |
| 4020-053038   |                       |                    | 1.9°             | 30               | 9,500                                                           | 2,450              | 0.06                            | 0.85                             | 5,500                                      | 920                | 0.03                            | 0.43                             | 4,250                                      | 480                | 0.017                           | 0.14                             |
| 4020-054038   |                       |                    |                  | 40               | 9,500                                                           | 2,450              | 0.05                            | 0.85                             | 5,500                                      | 920                | 0.025                           | 0.43                             | 4,250                                      | 480                | 0.015                           | 0.13                             |
| 4020-053058   |                       |                    | 2.9°             | 30               | 9,500                                                           | 2,450              | 0.07                            | 0.85                             | 5,500                                      | 920                | 0.035                           | 0.45                             | 4,250                                      | 480                | 0.017                           | 0.16                             |
| 4020-054258   |                       |                    |                  | 42               | 9,500                                                           | 2,450              | 0.06                            | 0.85                             | 5,500                                      | 920                | 0.03                            | 0.45                             | 4,250                                      | 480                | 0.016                           | 0.15                             |
| 4030-082008   | 3                     | R0.8               | 0.4°             | 20               | 6,500                                                           | 2,500              | 0.12                            | 1.06                             | 3,900                                      | 960                | 0.05                            | 0.58                             | 3,200                                      | 550                | 0.029                           | 0.25                             |
| 4030-082608   |                       |                    |                  | 26               | 6,300                                                           | 2,400              | 0.08                            | 1.04                             | 3,800                                      | 940                | 0.038                           | 0.56                             | 3,100                                      | 520                | 0.025                           | 0.22                             |
| 4030-083008   |                       |                    |                  | 30               | 6,300                                                           | 2,400              | 0.064                           | 1.01                             | 3,800                                      | 920                | 0.034                           | 0.55                             | 3,100                                      | 510                | 0.022                           | 0.21                             |
| 4030-083608   |                       |                    |                  | 36               | 6,300                                                           | 2,400              | 0.05                            | 1                                | 3,800                                      | 920                | 0.028                           | 0.52                             | 3,100                                      | 510                | 0.02                            | 0.19                             |
| 4030-084008   |                       |                    |                  | 40               | 6,300                                                           | 2,400              | 0.04                            | 0.98                             | 3,800                                      | 920                | 0.023                           | 0.51                             | 3,100                                      | 510                | 0.018                           | 0.17                             |



# HTNRS Milling Conditions

| WORK MATERIAL |                       |                    |                  |                  | PREHARDENED STEELS / HARDENED STEELS<br>NAK / SKD<br>(30~45HRC) |                    |                                 |                                  | HARDENED STEELS<br>SKD / SKT<br>(45~55HRC) |                    |                                 |                                  | HARDENED STEELS<br>SKD / SKH<br>(55~65HRC) |                    |                                 |                                  |
|---------------|-----------------------|--------------------|------------------|------------------|-----------------------------------------------------------------|--------------------|---------------------------------|----------------------------------|--------------------------------------------|--------------------|---------------------------------|----------------------------------|--------------------------------------------|--------------------|---------------------------------|----------------------------------|
| Model Number  | Outside Diameter (mm) | Corner Radius (mm) | Neck Taper Angle | Neck Length (mm) | Spindle Speed (min <sup>-1</sup> )                              | Feed Rate (mm/min) | a <sub>p</sub> Axial Depth (mm) | a <sub>e</sub> Radial Depth (mm) | Spindle Speed (min <sup>-1</sup> )         | Feed Rate (mm/min) | a <sub>p</sub> Axial Depth (mm) | a <sub>e</sub> Radial Depth (mm) | Spindle Speed (min <sup>-1</sup> )         | Feed Rate (mm/min) | a <sub>p</sub> Axial Depth (mm) | a <sub>e</sub> Radial Depth (mm) |
| 4030-082018   | 3                     | R0.8               | 0.9°             | 20               | 6,700                                                           | 2,600              | 0.13                            | 1.07                             | 4,000                                      | 1,000              | 0.065                           | 0.6                              | 3,300                                      | 590                | 0.034                           | 0.28                             |
| 4030-082518   |                       |                    |                  | 25               | 6,500                                                           | 2,500              | 0.1                             | 1.05                             | 3,900                                      | 960                | 0.05                            | 0.58                             | 3,200                                      | 550                | 0.029                           | 0.25                             |
| 4030-083018   |                       |                    |                  | 30               | 6,300                                                           | 2,400              | 0.072                           | 1.03                             | 3,800                                      | 920                | 0.038                           | 0.56                             | 3,100                                      | 510                | 0.024                           | 0.22                             |
| 4030-083518   |                       |                    |                  | 35               | 6,300                                                           | 2,400              | 0.064                           | 1.01                             | 3,800                                      | 920                | 0.034                           | 0.55                             | 3,100                                      | 510                | 0.022                           | 0.21                             |
| 4030-084018   |                       |                    |                  | 40               | 6,300                                                           | 2,400              | 0.056                           | 1                                | 3,800                                      | 920                | 0.03                            | 0.54                             | 3,100                                      | 510                | 0.021                           | 0.2                              |
| 4030-085018   |                       |                    |                  | 50               | 6,300                                                           | 2,400              | 0.04                            | 0.98                             | 3,800                                      | 920                | 0.023                           | 0.51                             | 3,100                                      | 510                | 0.018                           | 0.17                             |
| 4030-086018   |                       |                    |                  | 60               | 6,000                                                           | 2,300              | 0.024                           | 0.96                             | 3,600                                      | 870                | 0.015                           | 0.49                             | 2,900                                      | 470                | 0.015                           | 0.15                             |
| 4030-083028   |                       |                    | 1.4°             | 30               | 6,500                                                           | 2,500              | 0.09                            | 1.03                             | 3,900                                      | 960                | 0.045                           | 0.57                             | 3,200                                      | 550                | 0.03                            | 0.24                             |
| 4030-084028   |                       |                    |                  | 40               | 6,300                                                           | 2,400              | 0.06                            | 1.01                             | 3,800                                      | 920                | 0.035                           | 0.55                             | 3,100                                      | 510                | 0.02                            | 0.21                             |
| 4030-083038   |                       |                    | 1.9°             | 30               | 6,500                                                           | 2,500              | 0.1                             | 1.05                             | 3,900                                      | 960                | 0.05                            | 0.58                             | 3,200                                      | 550                | 0.03                            | 0.25                             |
| 4030-084038   |                       |                    |                  | 40               | 6,300                                                           | 2,400              | 0.07                            | 1.03                             | 3,800                                      | 920                | 0.04                            | 0.56                             | 3,100                                      | 510                | 0.025                           | 0.22                             |
| 4030-083358   |                       |                    | 2.9°             | 33               | 6,700                                                           | 2,500              | 0.12                            | 1.07                             | 3,900                                      | 1,000              | 0.06                            | 0.6                              | 3,200                                      | 590                | 0.03                            | 0.28                             |
| 4040-102508   | 4                     | R1                 | 0.4°             | 25               | 5,000                                                           | 2,600              | 0.17                            | 1.42                             | 3,000                                      | 1,000              | 0.085                           | 0.8                              | 2,450                                      | 600                | 0.045                           | 0.38                             |
| 4040-103008   |                       |                    |                  | 30               | 5,000                                                           | 2,600              | 0.13                            | 1.39                             | 2,900                                      | 960                | 0.065                           | 0.77                             | 2,400                                      | 540                | 0.038                           | 0.34                             |
| 4040-103508   |                       |                    |                  | 35               | 4,800                                                           | 2,450              | 0.09                            | 1.37                             | 2,900                                      | 920                | 0.048                           | 0.75                             | 2,350                                      | 480                | 0.032                           | 0.3                              |
| 4040-104008   |                       |                    |                  | 40               | 4,800                                                           | 2,450              | 0.08                            | 1.35                             | 2,900                                      | 920                | 0.043                           | 0.74                             | 2,350                                      | 480                | 0.03                            | 0.28                             |
| 4040-104508   |                       |                    |                  | 45               | 4,800                                                           | 2,450              | 0.07                            | 1.33                             | 2,900                                      | 920                | 0.038                           | 0.72                             | 2,350                                      | 480                | 0.028                           | 0.26                             |
| 4040-105008   |                       |                    |                  | 50               | 4,800                                                           | 2,450              | 0.06                            | 1.32                             | 2,900                                      | 920                | 0.034                           | 0.7                              | 2,350                                      | 480                | 0.026                           | 0.25                             |
| 4040-102018   |                       |                    | 0.9°             | 20               | 5,000                                                           | 2,600              | 0.19                            | 1.44                             | 3,000                                      | 1,000              | 0.095                           | 0.82                             | 2,450                                      | 600                | 0.048                           | 0.4                              |
| 4040-102518   |                       |                    |                  | 25               | 5,000                                                           | 2,600              | 0.17                            | 1.42                             | 3,000                                      | 1,000              | 0.085                           | 0.8                              | 2,450                                      | 600                | 0.045                           | 0.38                             |
| 4040-103018   |                       |                    |                  | 30               | 5,000                                                           | 2,600              | 0.15                            | 1.41                             | 3,000                                      | 1,000              | 0.076                           | 0.79                             | 2,450                                      | 600                | 0.042                           | 0.36                             |
| 4040-103518   |                       |                    |                  | 35               | 4,800                                                           | 2,450              | 0.12                            | 1.39                             | 2,900                                      | 960                | 0.062                           | 0.77                             | 2,400                                      | 540                | 0.037                           | 0.33                             |
| 4040-104018   |                       |                    |                  | 40               | 4,800                                                           | 2,450              | 0.09                            | 1.37                             | 2,900                                      | 920                | 0.048                           | 0.75                             | 2,350                                      | 480                | 0.032                           | 0.3                              |
| 4040-105018   |                       |                    |                  | 50               | 4,800                                                           | 2,450              | 0.08                            | 1.35                             | 2,900                                      | 920                | 0.043                           | 0.72                             | 2,350                                      | 480                | 0.029                           | 0.27                             |
| 4040-106018   |                       |                    |                  | 60               | 4,800                                                           | 2,450              | 0.06                            | 1.32                             | 2,900                                      | 920                | 0.034                           | 0.7                              | 2,350                                      | 480                | 0.026                           | 0.25                             |
| 4040-104928   |                       |                    | 1.4°             | 49               | 4,800                                                           | 2,500              | 0.1                             | 1.37                             | 2,900                                      | 960                | 0.05                            | 0.74                             | 2,350                                      | 540                | 0.035                           | 0.28                             |
| 4040-106028   |                       |                    |                  | 60               | 4,800                                                           | 2,500              | 0.08                            | 1.35                             | 2,900                                      | 960                | 0.04                            | 0.72                             | 2,350                                      | 540                | 0.03                            | 0.27                             |
| 4040-103038   |                       |                    | 1.9°             | 30               | 5,000                                                           | 2,600              | 0.15                            | 1.42                             | 3,000                                      | 1,000              | 0.08                            | 0.8                              | 2,450                                      | 600                | 0.045                           | 0.38                             |
| 4040-106738   |                       |                    |                  | 67               | 4,800                                                           | 2,500              | 0.12                            | 1.4                              | 2,900                                      | 960                | 0.05                            | 0.78                             | 2,350                                      | 540                | 0.03                            | 0.35                             |
| 4040-104558   |                       |                    | 2.9°             | 45               | 5,000                                                           | 2,600              | 0.15                            | 1.41                             | 3,000                                      | 1,000              | 0.08                            | 0.79                             | 2,450                                      | 600                | 0.045                           | 0.36                             |

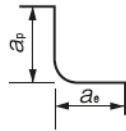
## HTNRS Milling Conditions

| WORK MATERIAL |                       |                    |                  |                  | PREHARDENED STEELS / HARDENED STEELS<br>NAK / SKD<br>(30~45HRC) |                    |                                 |                                  | HARDENED STEELS<br>SKD / SKT<br>(45~55HRC) |                    |                                 |                                  | HARDENED STEELS<br>SKD / SKH<br>(55~65HRC) |                    |                                 |                                  |
|---------------|-----------------------|--------------------|------------------|------------------|-----------------------------------------------------------------|--------------------|---------------------------------|----------------------------------|--------------------------------------------|--------------------|---------------------------------|----------------------------------|--------------------------------------------|--------------------|---------------------------------|----------------------------------|
| Model Number  | Outside Diameter (mm) | Corner Radius (mm) | Neck Taper Angle | Neck Length (mm) | Spindle Speed (min <sup>-1</sup> )                              | Feed Rate (mm/min) | a <sub>p</sub> Axial Depth (mm) | a <sub>e</sub> Radial Depth (mm) | Spindle Speed (min <sup>-1</sup> )         | Feed Rate (mm/min) | a <sub>p</sub> Axial Depth (mm) | a <sub>e</sub> Radial Depth (mm) | Spindle Speed (min <sup>-1</sup> )         | Feed Rate (mm/min) | a <sub>p</sub> Axial Depth (mm) | a <sub>e</sub> Radial Depth (mm) |
| 4060-152018   | 6                     | R1.5               | 0.9°             | 20               | 3,350                                                           | 2,600              | 0.28                            | 2.16                             | 2,000                                      | 1,000              | 0.14                            | 1.24                             | 1,650                                      | 600                | 0.072                           | 0.6                              |
| 4060-153018   |                       |                    |                  | 30               | 3,350                                                           | 2,600              | 0.28                            | 2.16                             | 2,000                                      | 1,000              | 0.14                            | 1.24                             | 1,650                                      | 600                | 0.072                           | 0.6                              |
| 4060-154018   |                       |                    |                  | 40               | 3,350                                                           | 2,600              | 0.26                            | 2.14                             | 2,000                                      | 1,000              | 0.131                           | 1.21                             | 1,650                                      | 600                | 0.068                           | 0.57                             |
| 4060-155018   |                       |                    |                  | 50               | 3,350                                                           | 2,600              | 0.2                             | 2.1                              | 2,000                                      | 1,000              | 0.103                           | 1.17                             | 1,650                                      | 600                | 0.058                           | 0.51                             |
| 4060-156018   |                       |                    |                  | 60               | 3,150                                                           | 2,400              | 0.14                            | 2.06                             | 1,900                                      | 920                | 0.075                           | 1.12                             | 1,550                                      | 510                | 0.048                           | 0.45                             |
| 4060-155128   |                       |                    | 1.4°             | 51               | 3,350                                                           | 2,600              | 0.2                             | 2.1                              | 2,000                                      | 1,000              | 0.1                             | 1.17                             | 1,650                                      | 600                | 0.058                           | 0.51                             |
| 4060-153938   |                       |                    | 1.9°             | 39               | 3,350                                                           | 2,600              | 0.26                            | 2.14                             | 2,000                                      | 1,000              | 0.13                            | 1.21                             | 1,650                                      | 600                | 0.068                           | 0.57                             |
| 4060-156938   |                       |                    |                  | 69               | 3,150                                                           | 2,400              | 0.14                            | 2.06                             | 1,900                                      | 920                | 0.075                           | 1.12                             | 1,550                                      | 510                | 0.048                           | 0.45                             |
| 4060-154758   |                       |                    | 2.9°             | 47               | 3,350                                                           | 2,600              | 0.26                            | 2.14                             | 2,000                                      | 1,000              | 0.13                            | 1.21                             | 1,650                                      | 600                | 0.068                           | 0.57                             |

Side Milling

a<sub>p</sub> : Axial Depth (mm)

a<sub>e</sub> : Radial Depth (mm)



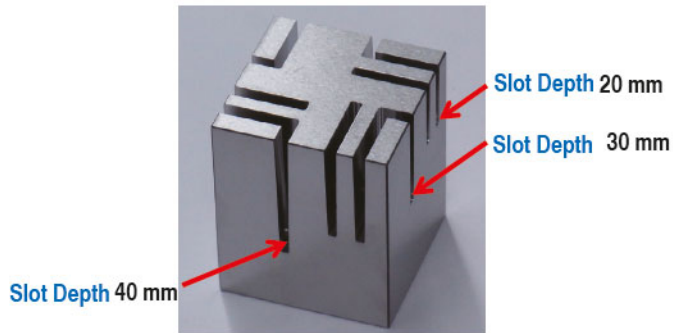
Note:

- Decrease both spindle speed and feed rate proportionally when the milling parameters exceed the machine's maximum spindle speed.
- Every coolant offers stable milling.

## Milling example of taper slotting

HTNRS  $\varnothing 2 \times \text{CR}0.5 \times \text{Neck Length } 20 \cdot 30 \cdot 40$

SKD61 (45HRC)



- Work Size :  $50 \times 50 \times 60 \text{ mm}$
- Inclined Angle :  $1^\circ$
- Slot Length : 27 mm (L Shape Slot)  
21 mm (Straight Slot)
- Slot Width : 2.6 mm (Bottom)
- Slot Depth : 20, 30, 40 mm
- Coolant : Water Soluble

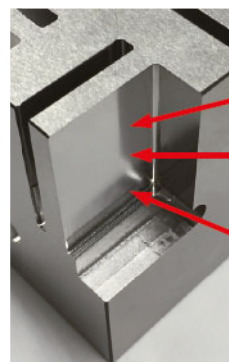
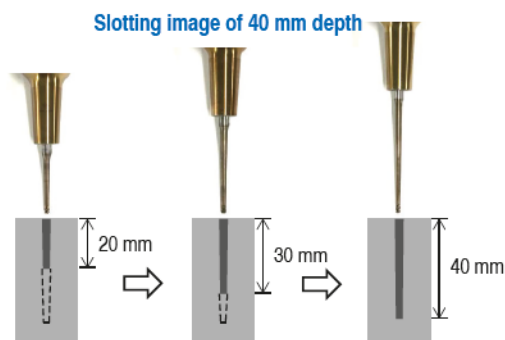
### ① Performance compared with straight neck type 20 mm Depth L shape slotting

| Process  | Tool              | Neck Shape<br>Helix Angle                        | Tool Size<br>(mm)                                                 | Spindle<br>Speed<br>( $\text{min}^{-1}$ ) | Feed<br>Rate<br>(mm/min) | $a_p$<br>Axial<br>Depth<br>(mm) | Cycle Time       |
|----------|-------------------|--------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------|--------------------------|---------------------------------|------------------|
| Roughing | HTNRS 4020-052018 | Taper Neck $0.9^\circ$<br>$45^\circ$ Helix Angle | $\varnothing 2 \times \text{CR}0.5 \times \text{Neck Length } 20$ | 9,500                                     | 2,450                    | 0.064                           | 20 min 18 sec    |
| Roughing | HLRS 4020-05-200  | Straight Neck<br>$30^\circ$ Helix Angle          | $\varnothing 2 \times \text{CR}0.5 \times$<br>Effective Length 20 | 7,000                                     | 800                      | 0.025                           | 1 h 30 min 9 sec |

**Taper neck 7 times more efficient in 20 mm depth slotting !**

### ② 40 mm Depth L shape slotting

| Process   | Tool              | Neck Shape<br>Helix Angle          | Tool Size<br>(mm)            | Spindle<br>Speed<br>(min <sup>-1</sup> ) | Feed<br>Rate<br>(mm/min) | a <sub>p</sub><br>Axial<br>Depth<br>(mm) | Cycle Time        |
|-----------|-------------------|------------------------------------|------------------------------|------------------------------------------|--------------------------|------------------------------------------|-------------------|
| Roughing  | HTNRS 4020-052018 | Taper Neck 0.9°<br>45° Helix Angle | Ø 2 × CR0.5 × Neck Length 20 | 9,500                                    | 2,450                    | 0.064                                    | 27 min 8 sec      |
| Roughing  | HTNRS 4020-053018 |                                    | Ø 2 × CR0.5 × Neck Length 30 | 9,500                                    | 2,450                    | 0.047                                    | 15 min 32 sec     |
| Roughing  | HTNRS 4020-054018 |                                    | Ø 2 × CR0.5 × Neck Length 40 | 9,000                                    | 2,300                    | 0.02                                     | 40 min 26 sec     |
| Finishing |                   |                                    |                              | 4,500                                    | 500                      | 0.0001<br>(Cusp Height)                  | 4 h 28 min 50 sec |



**Depth 20 mm**  
**Ra : 0.287  $\mu\text{m}$**

**Depth 30 mm**  
**Ra : 0.241  $\mu\text{m}$**

**Depth 40 mm**  
**Ra : 0.274  $\mu\text{m}$**

**40 mm slot depth roughing process completed in 1h 23 min ! Excellent surface finishing !**



### Advisory for Safe Use of UNIMAX Tungsten Carbide End Mills

Correct application and operation is strongly advised to avoid clogging, abrasion, etc., that could cause serious accidents or injuries. Ignition or sparks generated during milling could lead to fire or extreme damage to the work piece. End Mills are made with very sharp cutting edges and must be handled with extra care.

- \* Never touch the cutting edge with your bare hands, as this could cause serious injury. Special caution is required when opening the package.
- \* Dropping the tool could cause breakage or flying debris, leading to serious injury.
- \* During milling, unexpected impact or shock on the tool could cause breakage or flying debris. Ensure to use protective items such as safety glasses and a face guard.
- \* For best results, fine parameter adjustment may be required, depending on the materials; milling shape and strategy; machine rigidity and spindle capability.
- \* Use a machine that has high rigidity and generates a low level of vibration.
- \* Do not use flammable cutting oils.

#### Advisory for regrinding UNIMAX Tungsten Carbide End Mills

- \* Never regrind the tool without wearing safety glasses and a face guard.



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